

Darrin Bright

✉ darriebright@gmail.com 🌐 Portfolio 📄 GitHub 🎓 Scholar 🔗 LinkedIn

Research Interests

Medical Image Analysis | 3D Vision | Multimodal Learning

Education

Vellore Institute of Technology (VIT)

Integrated Master of Technology in Software Engineering

Vellore, India

2022 – 2027 (Expected)

CGPA: 8.81/10.0

Relevant Coursework: Artificial Intelligence, Machine Learning, Natural Language Processing, Data Mining, Database Management Systems, Python Programming

Publications

- [1] **D. Bright**, R. Raj, K. Keisham. “PortionNet: Distilling 3D Geometric Knowledge for Food Nutrition Estimation.” *Accepted at Conference on Vision and Intelligent Systems (CVIS)*, 2025. [[Paper](#)][[Code](#)]

Research Experience

Nanyang Technological University

Research Intern | Advisor: Dr. Si Yong Yeo

Singapore

July 2026 – Present

- Developing semi-supervised deep learning models for coronary artery segmentation from X-ray angiography under limited annotation settings.

Simon Fraser University

Research Intern | Advisor: Dr. Ghassan Hamarneh

Vancouver, Canada

December 2025 – August 2026

- Built a 3D detection pipeline for mitochondria-ER contact sites in STED microscopy volumes.
- Generated training data from both real STED acquisitions and a synthetic pipeline that renders parametric organelle masks into realistic STED-like images via the pySTED simulator.

Indian Institute of Technology, Kharagpur

Research Intern | Advisor: Dr. Debanjan Das

Kharagpur, India

May 2026 – July 2026

- Developed GraphUNet, encoding beat-level ECG and PPG morphology as graphs within a U-Net for cuffless blood pressure estimation.
- Benchmarked GraphUNet against published state-of-the-art baselines on VitalDB and PulseDB, achieving competitive estimation performance with interpretable beat-level graph representations.

Indian Institute of Technology, Hyderabad

Research Intern | Advisor: Dr. C. Krishna Mohan

Hyderabad, India

January 2026 – June 2026

- Built a federated continual learning framework combining a global synthetic-data anchor with V-Information selective re-distillation, outperforming FedCIL and FedLwF-2T across five MedMNIST modalities.

Vellore Institute of Technology

Undergraduate Researcher

Vellore, India

July 2025 – June 2026

SPHINX-Net: A Lightweight Histology-Aware Framework for OSCC Detection

Advisor: Dr. Bharanidharan N

- Developed SPHINX-Net, a lightweight hybrid framework with histology-aware multi-view contrastive pretraining, enabling strong patch-level OSCC detection.
- Designed a spectral-aware recalibration block that reweights channels by frequency-domain energy to target the nuclear-texture irregularity central to tumour grading.
- Introduced a prototype-graph head that resolves ambiguous patches through mini-batch graph context while enforcing compact class geometry, without whole-slide-image graphs.

PortionNet: Distilling 3D Geometric Knowledge for Food Nutrition Estimation

Advisor: Dr. Kanchan Keisham

- Developed PortionNet, a cross-modal knowledge distillation framework enabling RGB models to learn 3D geometric features from point clouds during training, eliminating depth sensor requirements at inference.
- Designed a dual-mode training strategy with a lightweight RGB-to-Geometry Adapter that learns to generate pseudo-3D features for accurate geometric reasoning from a standard RGB image.
- Achieved state-of-the-art performance on MetaFood3D and SimpleFood45 datasets, demonstrating effective generalization.

Indian Institute of Technology, Jodhpur

Research Intern | Advisor: Dr. Deepak Mishra

Jodhpur, India

May 2025 – July 2025

Self-Supervised Medical Image Segmentation with Vector Quantization

- Developed a hierarchical vector quantization framework that addresses limited labeled data in medical imaging by learning discrete representations from unlabeled ISIC 2018 images and using them as structured regularizers during segmentation fine-tuning on the PH2 dataset.
- Designed three-stage training pipeline integrating VQ-VAE, SimCLR contrastive learning and segmentation fine-tuning, with multi-layer quantization capturing features across semantic scales.
- Achieved 79.76% Dice, a +1.38% consistent improvement over baseline without VQ across multiple runs.

Ablation Study: Vector Quantization Integration in Transformer-based Segmentation

- Conducted a 13-experiment ablation study on Pascal VOC 2012 dataset investigating vector quantization integration with Swin Transformer encoders, testing codebook configurations, fusion mechanisms and initialization strategies.
- Achieved 74.61% Dice score with optimized vector quantization configuration, demonstrating +7.34% improvement over baseline CNN and +3.91% over vanilla Swin Transformer, validating VQ effectiveness for semantic segmentation.

Technical Skills

Programming: Python, SQL

ML/DL Frameworks: PyTorch, TensorFlow, MONAI

Computer Vision: OpenCV, PointNet, Albumentations, Torchvision

Libraries & Tools: NumPy, Pandas, Matplotlib, SciPy, Scikit-learn

Honors & Awards

- **Runner-up**, Startup Demo Day, Vellore Institute of Technology, 2025
- **Second Runner-up**, HackWar Hackathon, Vellore Institute of Technology, 2025
- **Winner**, Project 2039 Hackathon, Vellore Institute of Technology, 2024
- **Winner**, Alphaforge Ideathon, Vellore Institute of Technology, 2024
- **Runner-up**, Biomimicry Innovation Challenge, Vellore Institute of Technology, 2024

Service & Leadership

IEEE Robotics and Automation Society, VIT

Chairperson

Vellore, India

December 2024 – December 2025

- Led a 500+ member student chapter, organizing technical workshops, hackathons and research talks, while mentoring junior students through hands-on sessions on ML fundamentals and Generative AI frameworks (LLMs, LangChain).

National Service Scheme

Volunteer

Vellore, India

January 2023 – January 2024

- Conducted career guidance sessions for government school students on educational pathways and professional opportunities.